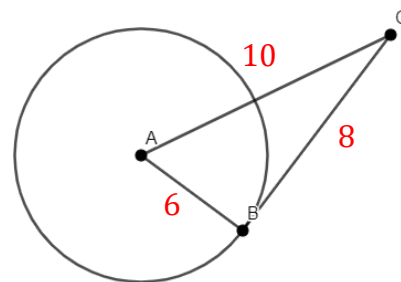


Geometry
Unit 12
Lesson 5 Homework

Name **Answer Key** _____
Date _____
Period _____

1. Circle A is shown with point C outside the circle, where point B is on the circle, $AC = 10$, $CB = 8$, and $BA = 6$. Verify if $\overline{BC}$ is tangent to circle A using the Pythagorean Theorem.

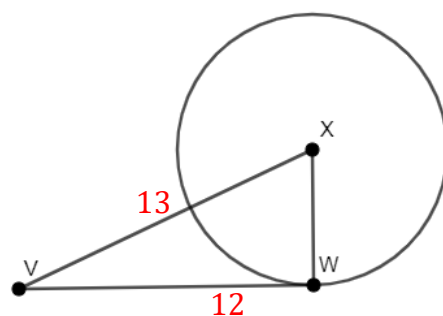
$$\begin{aligned} 6^2 + 8^2 &= 10^2 \\ 36 + 64 &= 100 \\ 100 + 100 &\checkmark \Rightarrow \text{TANGENT} \end{aligned}$$



Circle X on the right has $\overline{VW}$ as a tangent segment where W is the point of tangency. $VX = 13$ and $VW = 12$.

2. What is the length of $\overline{XW}$?

$$\begin{aligned} 12^2 + (XW)^2 &= 13^2 \\ (XW)^2 &= 25 \\ \boxed{XW} &= 5 \end{aligned}$$



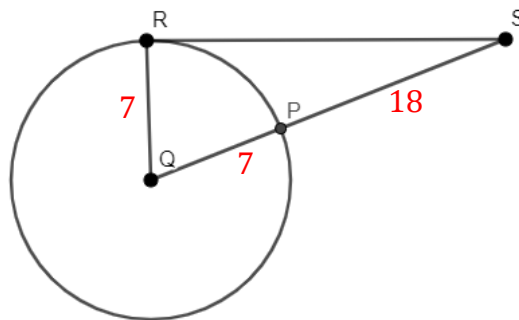
3. What is the area of $\triangle XWV$?

$$\frac{1}{2}(12)(5) = \boxed{30}$$

In Circle Q , $\overline{RS}$ is tangent to the circle at point R , $QR = 7$, and $PS = 18$.

4. What is the length of $\overline{RS}$?

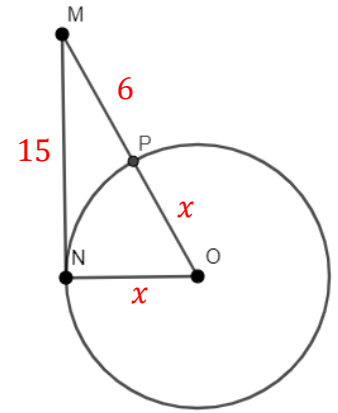
$$\begin{aligned} (RS)^2 + 7^2 &= (7 + 18)^2 \\ (RS)^2 + 49 &= 625 \\ (RS)^2 &= 576 \\ \boxed{RS} &= 24 \end{aligned}$$



5. What is the area of $\triangle QRS$?

$$\frac{1}{2}(7)(24) = \boxed{84}$$

In Circle O , $\overline{MN}$ is a tangent segment whose point of tangency is point N . $MN = 15$ and $MP = 6$.



6. What is the length of $\overline{ON}$?

$$\begin{aligned}x^2 + 15^2 &= (6 + x)^2 \\x^2 + 225 &= 36 + 12x + x^2 \\12x &= 189 \\x &= \boxed{ON = 15.75}\end{aligned}$$

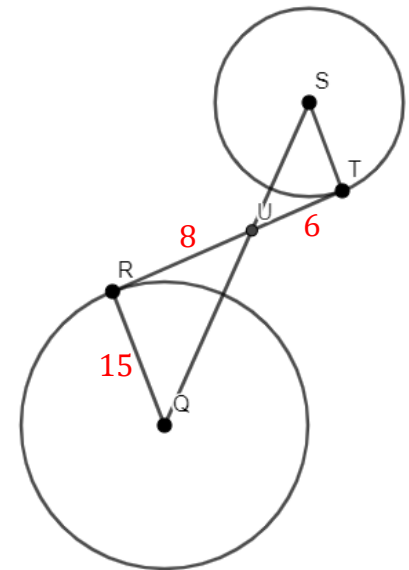
7. What is the length of $\overline{MO}$?

$$6 + (15.75) = \boxed{MO = 21.75}$$

8. What is the perimeter of $\triangle MNO$?

$$15 + 15.75 + 21.75 = \boxed{52.5}$$

In the figure on the right, $\overline{RT}$ is a common tangent segment of both Circle Q and Circle S . $QR = 15$, $RU = 8$, and $TU = 6$.



9. What is the length of $\overline{ST}$?

$$\begin{aligned}\frac{8}{6} &= \frac{15}{ST} \\8(ST) &= (6)(15) \\\boxed{ST} &= 11.25\end{aligned}$$

10. What is the length of $\overline{QS}$?

$$\begin{aligned}\overline{QU} &= \sqrt{8^2 + 15^2} = 17 \\\overline{US} &= \sqrt{6^2 + (11.25)^2} = 12.75 \\\overline{QS} &= 17 + 12.75 \\\boxed{QS} &= 29.75\end{aligned}$$